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Centro de Investigación en Computación

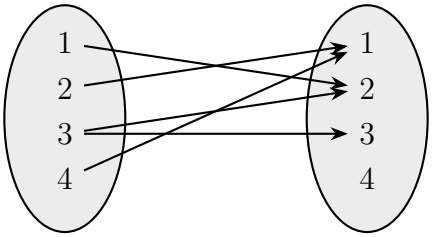
Mathematics for Computer Science

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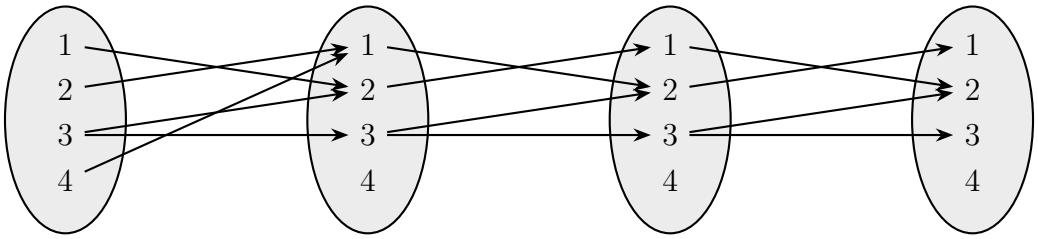
Homework 3

Given $A = \{1, 2, 3, 4\}$, $R \subseteq A \times A$, $R = \{(1, 2), (2, 1), (3, 2), (3, 3), (4, 1)\}$.
Find:
$R^3 = R \circ R \circ R \subseteq A \times A$
Represent the initial graph R and the final graph R^3 as binary graphs.

Initial graph for R



Final graph for R^3



Then, we have $R^3 = \{(1, 2), (2, 1), (3, 2), (3, 1), (3, 3), (4, 1)\}$.